MILITARY SPECIFICATION

MICROCIRCUITS, LINEAR, INSTRUMENTATION
AMPLIFIERS, MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, instrumentation amplifiers. Two product assurance classes and a choice of case outlines and lead finishes are provided and are reflected in the complete part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510.

1.2.1 Device type. The device type shall be as follows:

Device type	Circuit
01	Instrumentation amplifier

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Outline letter	Case outline (see MIL-M-38510, appendix C)
E	D-2 (16-lead, 1/4" x 7/8"), dual-in-line package

1.3 Absolute maximum ratings.

Positive supply voltage (V_{CC}) to common	+18 V dc
Negative supply voltage (V_{EE}) to common	-18 V dc
Differential input voltage range pins 1 and 2	±36 V
Common mode input voltage range	±18 V
Output short-circuit duration	1/
Junction temperature (T_J)	+175°C
Storage temperature range	-65°C to +150°C
Lead temperature (soldering, 60 seconds)	+300°C
Power dissipation (P_D)	280 mW

1.4 Recommended operating conditions.

Positive supply voltage range (V_{CC})	+10.8 V dc to +16.5 V dc
Negative supply voltage range (V_{EE})	-16.5 V dc to -10.8 V dc
Difference mode input voltage (V_{DM})	-10/A V dc to +10/A V dc 2/
Common mode input voltage (V_{CM})	(10-(V_{DM}/2)) 2/
Ambient operating temperature range (T_A)	-55°C to +125°C

1/ The duration of the output short circuit for any combination of supply voltage, output short-circuit current, and temperature shall not cause the junction temperature to exceed its maximum rating.
2/ A = programmable device gain.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1.5 Power and thermal characteristics.

Package	Case outline	Maximum allowable power dissipation	Maximum θ_{JC}	Maximum θ_{JA}
16-lead DIP	E	280 mW at T_A = +125°C	25°C/W	90°C/W

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standard. The following specification and standard form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of the specification and standard required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets, or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

3.2.1 Functional diagram and terminal connections. The functional diagram and terminal connections shall be as specified on figure 1.

3.2.2 Functional schematic circuit. The functional schematic circuit shall be as specified on figure 2.

3.2.3 Schematic circuits. The schematic circuits shall be submitted to the preparing activity prior to inclusion of manufacturer's device in this specification and shall be submitted to the qualifying activity and agent activity (DESC-ECS) as a prerequisite for qualification. All qualified manufacturer's schematics shall be maintained by the agent activity and available upon request.

3.2.4 Case outline. The case outline shall be as specified in 1.2.3 herein.

3.2.5 Package and sealing material. Package and sealing material shall be in accordance with MIL-M-38510.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 and 6.4 herein.

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in table I, and unless otherwise stated, apply over the full recommended ambient operating temperature range for supply voltages from ±10.8 V dc to ±16.5 V dc.

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TABLE I. Electrical performance characteristics.

Characteristic	Symbol	Conditions V_{CC} = ±15 V, R_L = 2 kΩ, unless otherwise specified (see 3.4 and figure 3)		Limits Min	Limits Max	Unit
Quiescent current (+)	$I_{CC}(+)$	V_{IN} = 0 V; A = 1			5.0	mA
Quiescent current (-)	$I_{EE}(-)$	V_{IN} = 0 V; A = 1		-5.0		mA
Output short circuit current	$I_{OS}(+)$	V_{IN} = 0 V; A = 1		10	85	mA
	$I_{OS}(-)$	V_{IN} = 0 V; A = 1		-85	-10	mA
Input offset voltage	V_{IOS}	V_{IN} = 0 V; A = 1, 1000 1/	T_A = +25°C	-100	100	µV
			T_A = +125°C	-300	300	µV
			T_A = -55°C	-260	260	µV
Input offset voltage temperature sensitivity	dV_{IOS}/dT	V_{IN} = 0 V; A = 1, 1000 -55°C ≤ T_A ≤ +125°C 1/		-2	2	µV/°C
Output offset voltage	V_{OOS}	V_{IN} = 0 V; A = 1, 1000 1/	T_A = +25°C	-3	3	mV
			T_A = +125°C	-8	8	mV
			T_A = -55°C	-7	7	mV
Output offset voltage temperature sensitivity	dV_{OOS}/dT	V_{IN} = 0 V; A = 1, 1000 -55°C ≤ T_A ≤ +125°C 1/		-50	50	µV/°C
Adjustment for input offset	V_{IOS} Adj	V_{IN} = 0 V; A = 1000 T_A = +25°C 1/		-100	100	µV
Adjustment for output offset	V_{OOS} Adj	V_{IN} = 0 V; A = 1 T_A = +25°C 1/		-3	3	mV
Input bias current (+)	$I_{IB}(+)$	V_{IN} = 0 V; A = 1	T_A = +25°C	-50	50	nA
			-55°C ≤ T_A ≤ +125°C	-70	70	nA

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Characteristic	Symbol	Conditions V_{CC} = ±15 V, R_L = 2 kΩ, unless otherwise specified (see 3.4 and figure 3)		Limits Min	Limits Max	Unit
Input bias current (-)	$I_{IB}(-)$	V_{IN} = 0 V; A = 1	T_A = +25°C	-50	50	nA
			-55°C ≤ T_A ≤ +125°C	-70	70	nA
Input offset current	I_{IO}	T_A = +25°C		-35	35	nA
		-55°C ≤ T_A ≤ +125°C		-50	50	nA
Power supply rejection ratio (+) RTI	P_{SRR}	10.8 V dc ≤ V_{CC} ≤ 16.5 V dc, -16.5 V dc ≤ V_{EE} ≤ -10.8 V dc 2/	A = 1	75		dB
			A = 10	95		dB
			A = 100	105		dB
			A = 1000	110		dB
Common mode rejection ratio	C_{MRR}	0 V dc ≤ V_{IN} ≤ 10 V dc, -10 V dc ≤ V_{IN} ≤ 0 V dc	A = 1	70		dB
			A = 10	90		dB
			A = 100	100		dB
			A = 1000	110		dB
Gain error	A_E	T_A = +25°C	A = 1	-0.05	0.05	%
			A = 10	-0.25	0.25	%
			A = 100	-0.5	0.5	%
			A = 1000	-2.0	2.0	%
Gain error temperature sensitivity	dA_E/dT	-55°C ≤ T_A ≤ +125°C	A = 1	-5	5	ppm/°C
			A = 10	-10	10	ppm/°C
			A = 100	-25	25	ppm/°C
			A = 1000	-50	50	ppm/°C
Output voltage swing (minimum)	V_{OP}	R_L = 2 kΩ	V_{IN} = +15 V	10	15	V
			V_{IN} = -15 V	-15	-10	V

See footnotes at end of table.

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TABLE I. Electrical performance characteristics - Continued.

Characteristic	Symbol	Conditions V_{CC} = ±15 V, R_L = 2 kΩ, unless otherwise specified (see 3.4 and figure 3)		Limits Min	Limits Max	Unit
Settling time (to .01%)	ts	T_A = +25°C See figure 4	A = 1		15	µs
			A = 10		15	µs
			A = 100		15	µs
			A = 1000		75	µs
Slew rate (+)	SR(+)	T_A = +25°C	A = 1	3.5		V/µs
			A = 1000	0.5		V/µs
Slew rate (-)	SR(-)	T_A = +25°C	A = 1	3.5		V/µs
			A = 1000	0.5		V/µs
Input noise voltage density	En	T_A = +25°C; Gain = 1 See figure 5; 1,000 Hz			350	$nV/\sqrt{Hz}$
Input noise current density	In	T_A = +25°C; Gain = 1 See figure 5; 1,000 Hz			10	$pA/\sqrt{Hz}$
Low frequency input noise V	Enpp	T_A = +25°C; Gain = 1 See figure 6; 0.1 Hz ≤ BW ≤ 10 Hz			15	µVp-p
Differential input resistance	RiD	T_A = +25°C Information only		1		GΩ
Differential input capacitance	CiD	T_A = +25°C Information only			10	pF
Common mode input resistance	RiCM	T_A = +25°C Information only		1		GΩ
Common mode input capacitance	CiCM	T_A = +25°C Information only			10	pF
Gain nonlinearity	GLE	Information only 3/	A = 1	-0.01	0.01	%FSR 4/
			A = 10	-0.01	0.01	%FSR 4/
			A = 100	-0.01	0.01	%FSR 4/
			A = 1000	-0.01	0.01	%FSR 4/

1/ Adjustable to zero.
2/ V_{CC} and V_{EE} are tracking supplies (V_{CC} = $-V_{EE}$).
3/ Gain nonlinearity is specified as the maximum deviation from a line defined using the least squares method over the range of the device at that gain.
4/ FSR = full scale range (corresponds to full scale frequency or full scale input voltage).

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3.5 Electrical test requirements. The electrical test requirements for each device class shall be the subgroups specified in table II. The electrical tests for each subgroup are described in table III.

3.6 Marking. Marking shall be in accordance with MIL-M-38510 and 1.2 herein.

3.6.1 Serialization. All class S devices shall be serialized in accordance with MIL-M-38510.

3.6.2 Correctness of indexing and marking. All devices shall be subjected to the final electrical tests specified in table II after part marking to verify that they are correctly indexed and identified by part number. Optionally, an approved electrical test may be devised especially for this requirement.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 49 (see MIL-M-38510, appendix E).

TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	Subgroups (see table III) 1/	
	Class S devices	Class B devices
Interim electrical parameters (method 5004)	1	1
Final electrical test parameters (method 5004)	1, 2, 3, 4	1, 2, 3, 4
Group A test requirements (method 5005)	1, 2, 3, 4, 5, 6, 7	1, 2, 3, 4, 5, 6, 7
Group C end-point and group B class S, electrical parameters (method 5005)	1, 2, 3, and table IV delta limits	1 and table IV delta limits
Group D end-point electrical parameters (method 5005)	1, 2, 3, and table IV end-point limits	1 and table IV end-point limits
Additional electrical subgroups for group C periodic inspection	N/A	---

1/ PDA applies to subgroup 1 (see 4.2c).

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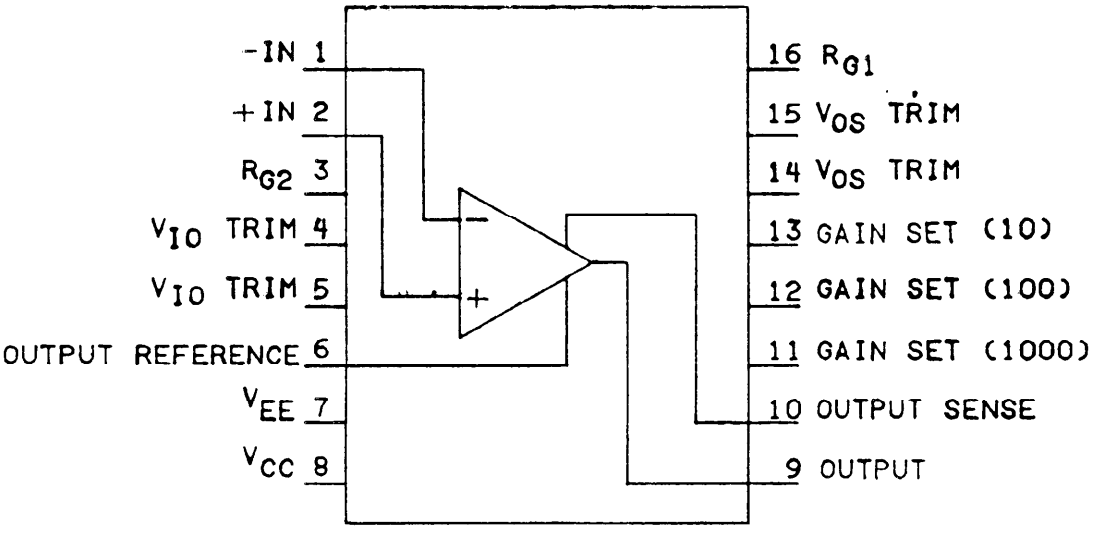
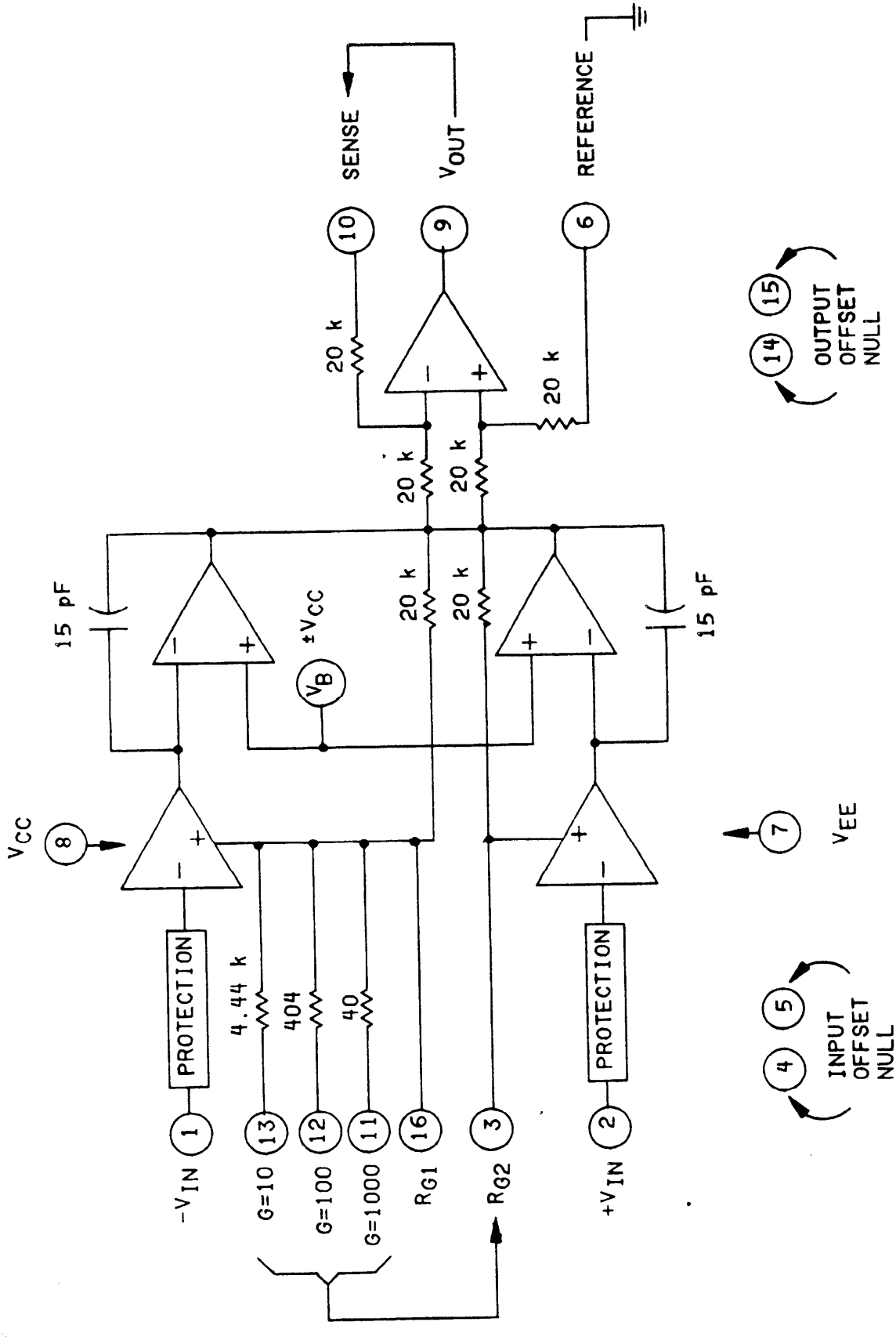


FIGURE 1. Functional diagram and terminal connection.

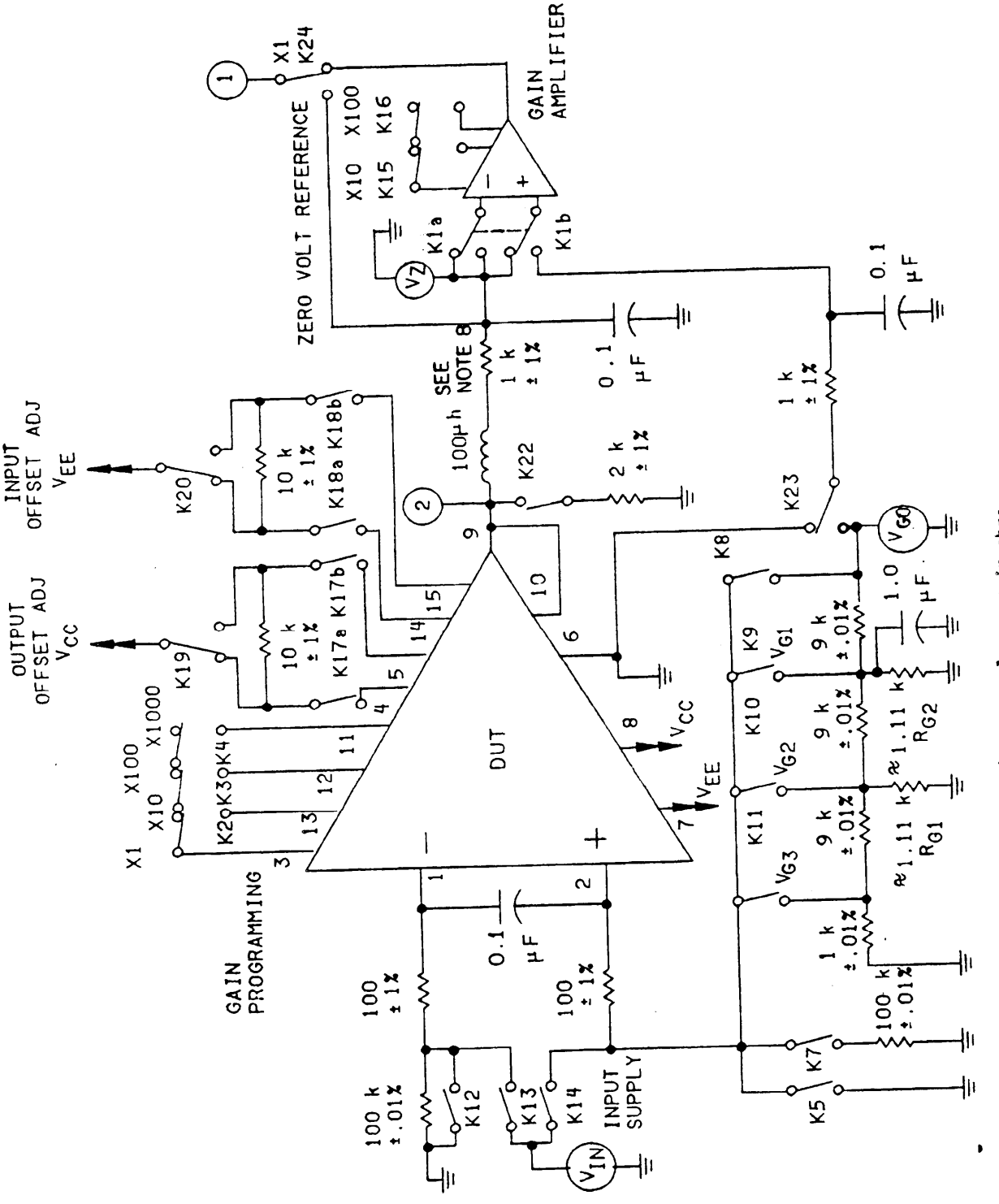
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NOTE: Resistance values are in ohms.

FIGURE 2. Functional schematic circuit.

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NOTE: Resistance values are in ohms.

FIGURE 3. Test circuit for static tests.

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NOTES:
1. All resistor tolerances are as shown and all capacitors are ±10 percent, unless otherwise specified.
2. Programmable voltages V_{G1}, V_{G2} and V_{G3} are adjusted to meet the tne following minimum requirements:

$$0.0985 \leq \frac{V_{G3}}{V_{G2}} \leq 0.1015$$

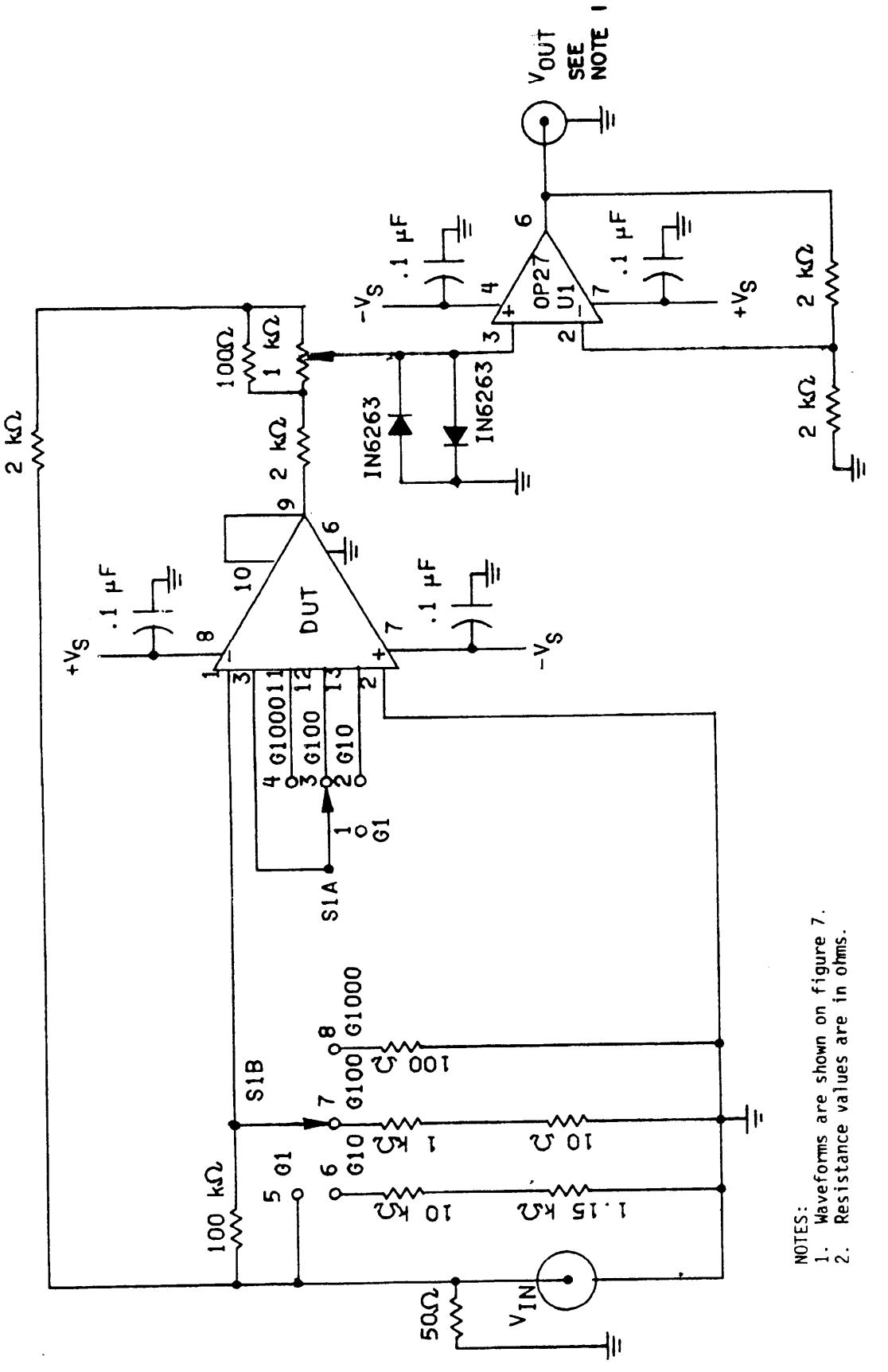
   a. Select a value of R_{G1} for $0.09975 \leq \frac{V_{G2}}{V_{G1}} \leq 0.10025$

   b. Select a value of R_{G2} for $0.09975 \leq \frac{V_{G1}}{V_{G0}} \leq 0.10025$

3. Precautions snall be taken to prevent damage to tne DUT during insertion into the socket during changes of relay state (i.e., disable voltage supplies, current limit $\pm V_{CC}$, etc.).
4. Proper wiring procedures shall be followed to prevent unwanted coupling and oscillations.
5. Adequate settling time shall be allowed such that each parameter has settled to within 5 percent of its final value.
6. All relays are shown in the normal deenergized state.
7. Any oscillation greater than 300 mV pk-pk in amplitude shall be cause for device failure.
8. A 100-uH inductor may be used to reduce output oscillations in the test circuit.
9. The CMRR, referenced to the input, of the gain amplifier shall be equal to or greater than 90 dB.

FIGURE 3. Test circuit for static tests - Continued.

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NOTES:
1. Waveforms are shown on figure 7.
2. Resistance values are in ohms.

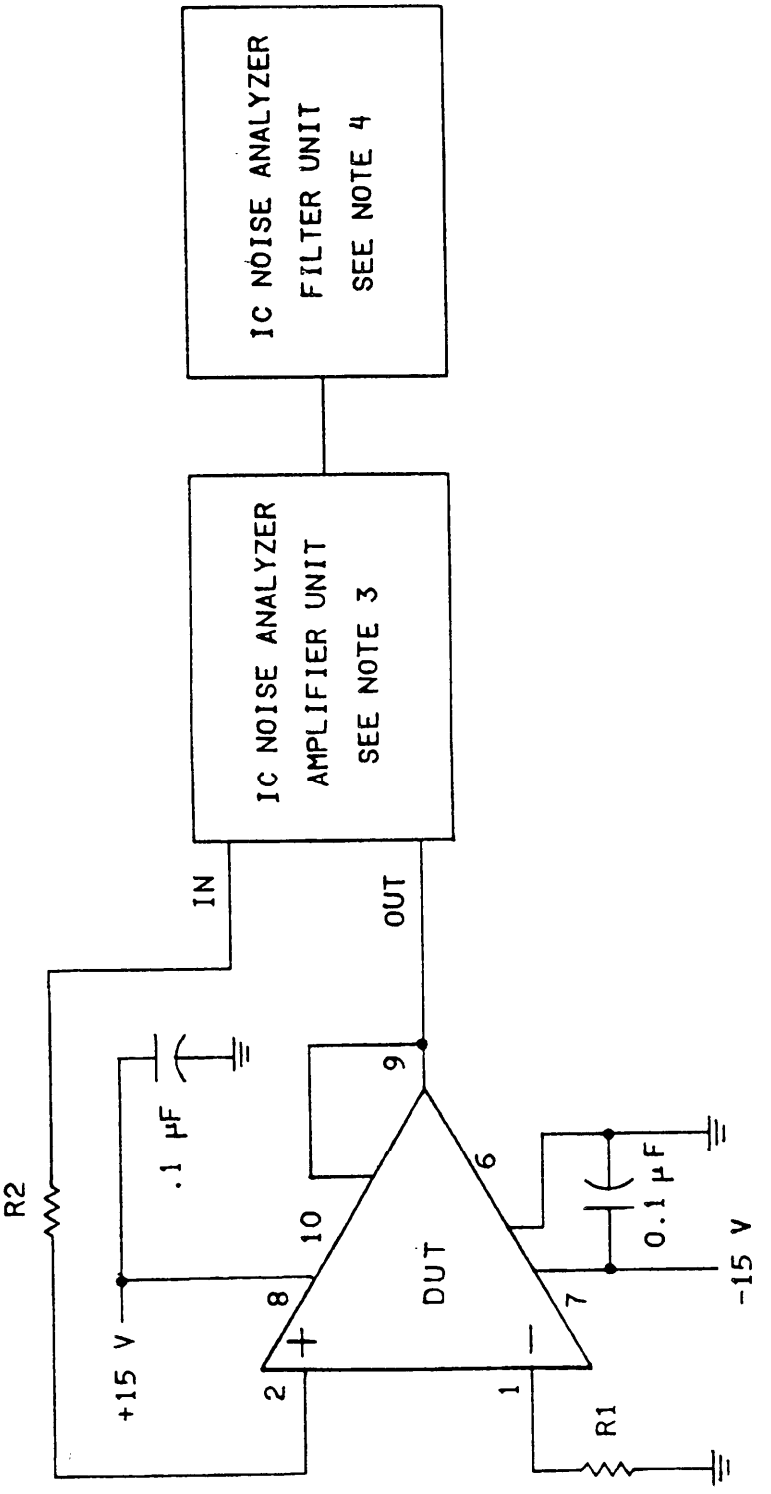
FIGURE 4. Test circuit for settling time.

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NOTES:
1. All resistors are metal film with a ±0.1 percent tolerance, and all capacitors are ±10 percent unless otherwise specified.
2. The potentiometer is a 1 turn device.
3. D1 and D2 are IN6263 or equivalent types.
4. U1 is an OP27 or equivalent type.
5. V_{IN} shall be a recurrent square wave with a 10 volt amplitude, a 50 percent duty cycle, and a frequency of 1 kHz. The signal generator shall have a low noise output.
6. Input and output waveforms are shown on figure 7.
7. The trimpot shall be adjusted until horizontal traces converge.
8. The settling time is defined to be the time required for the output to reach and maintain 0.01 percent of the final value. This corresponds to 1.0 mV.
9. Precautions shall be taken to prevent damage to the DUT during insertion into the socket with the power applied.
10. Proper wiring procedures shall be followed to prevent unwanted coupling, noise, and oscillations.

FIGURE 4. Test circuit for settling time - Continued.

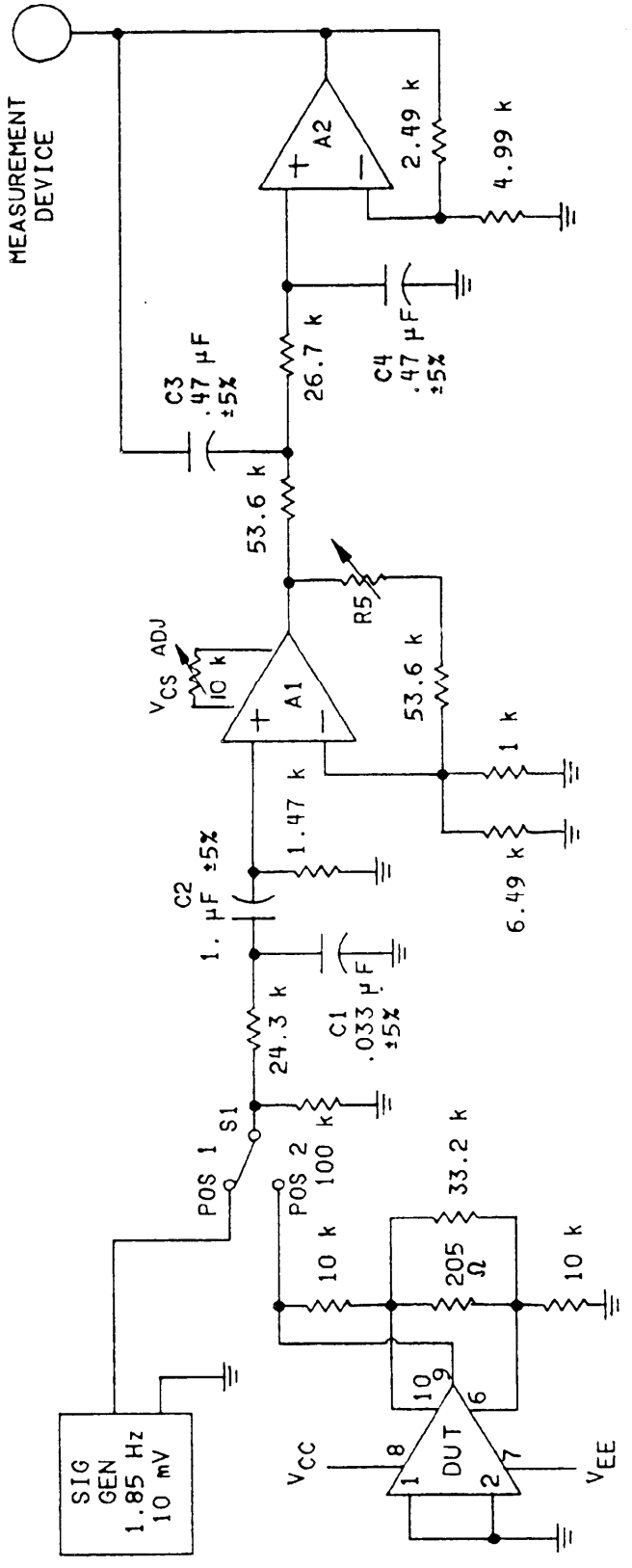
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NOTES:
1. Input noise voltage density (En) test: R1 = R2 = 0Ω.
   Input noise current density (In) test: R1 = R2 = 10 MΩ.
2. All resistors are metal film and ±1 percent tolerance. Capacitors are in microfarads and are ±10% tolerance.
3. Quan-Tech model 2283 or equivalent.
4. Quan-Tech model 2181 or equivalent.
5. Read En directly from Quan-Tech meters.
6. $In = \frac{En}{\sqrt{2}\,(10^7)}$

FIGURE 5. Noise density test circuit.

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NOTES:
1. Trim R5 to provide filter gain of 100 at 1.05 Hz, with S1 in position 1.
2. Decouple power supplies with 0.1 µF ceramic capacitors (not shown).
3. C3 must be the larger of the two capacitors (C3 and C4).
4. C1, C2, C3, C4 are polypropylene wrap end fill capacitors.
5. All resistors are ±1% metal film RN55C and values are in ohms.
6. A1 and A2 are AD547 and AD517, respectively, or equivalent and must have low noise (< 2 μV_{p-p}) and no popcorn noise.
7. Measure the peak-peak noise with Tektronix 1720 or equivalent, with S1 in position 2. Set the time base for a 10-second full sweep.

FIGURE 6. Low frequency test circuit.

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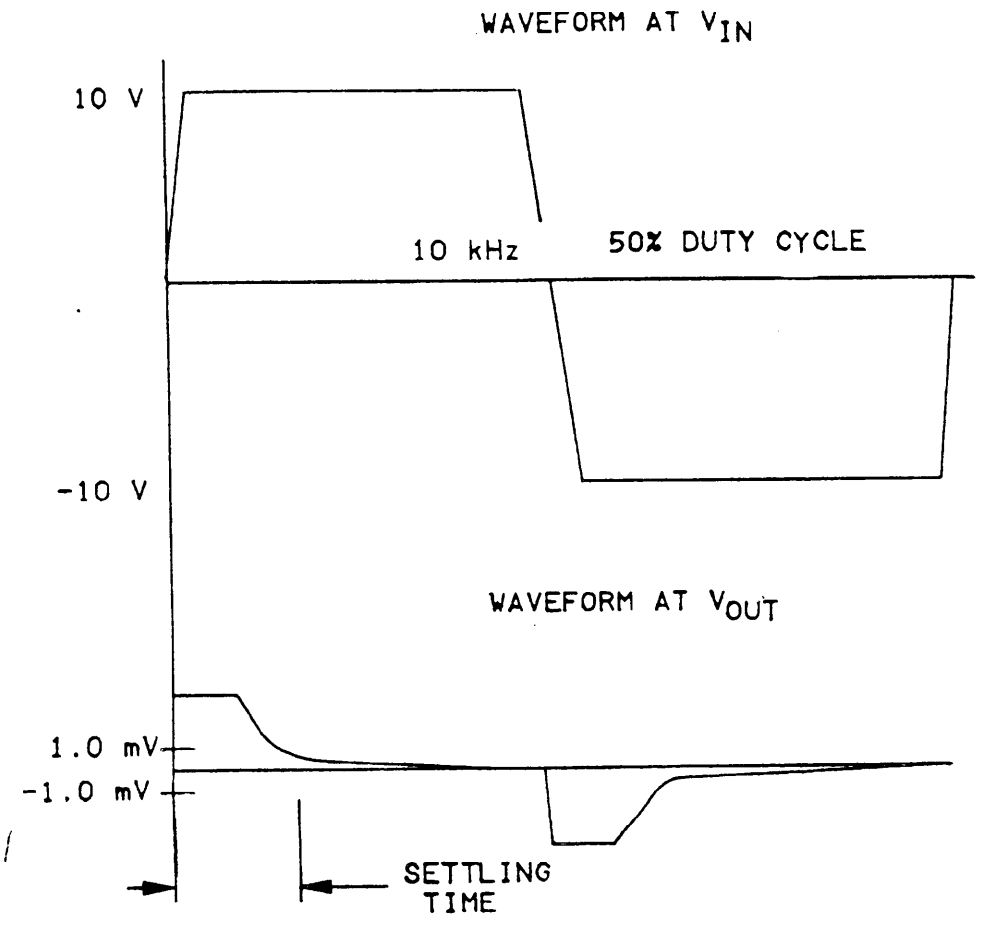
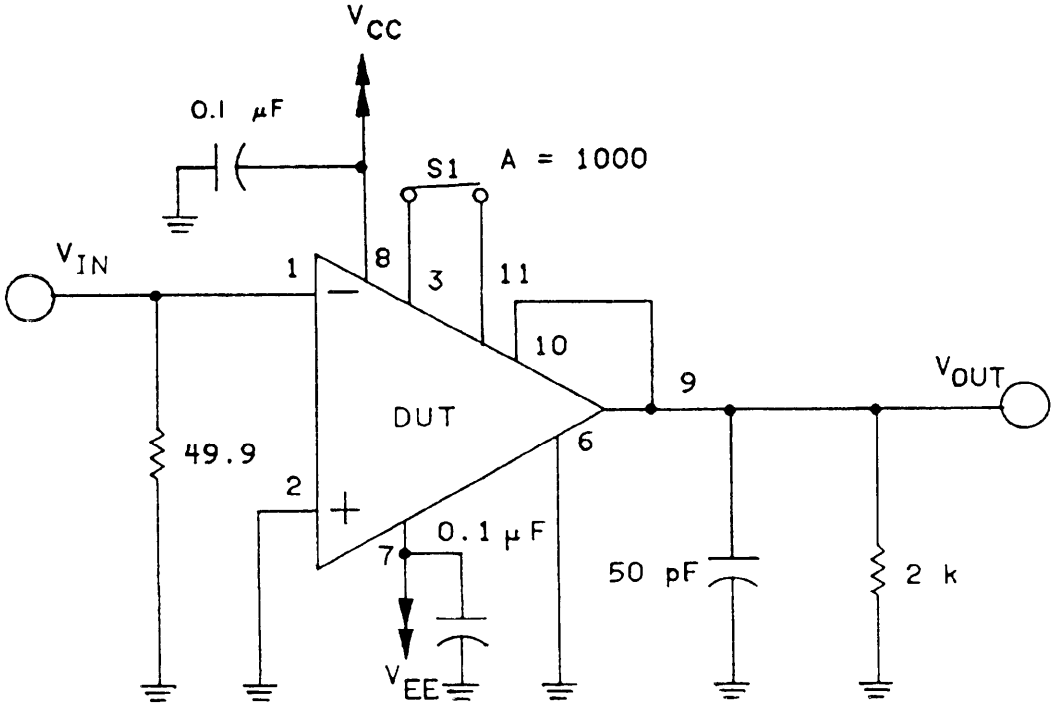


FIGURE 7. Settling time waveforms.

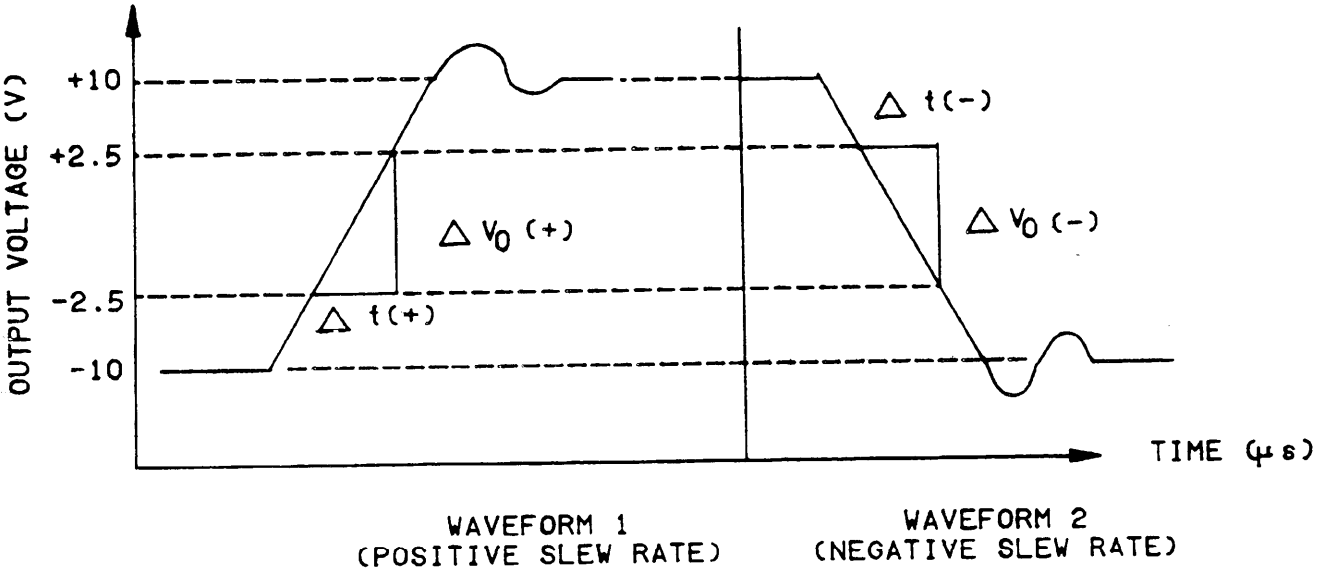
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NOTES:
1. Resistor values are in ohms with a ±1.0 percent tolerance and capacitors are ±10 percent tolerance.
2. This capacitance includes the actual measured value with stray and wire capacitance.
3. Precautions shall be taken to prevent damage to the DUT during insertion into the socket and in applying power.
4. Pulse input and output characteristics are shown on figure 9.

FIGURE 8. Test circuit for slew rate.

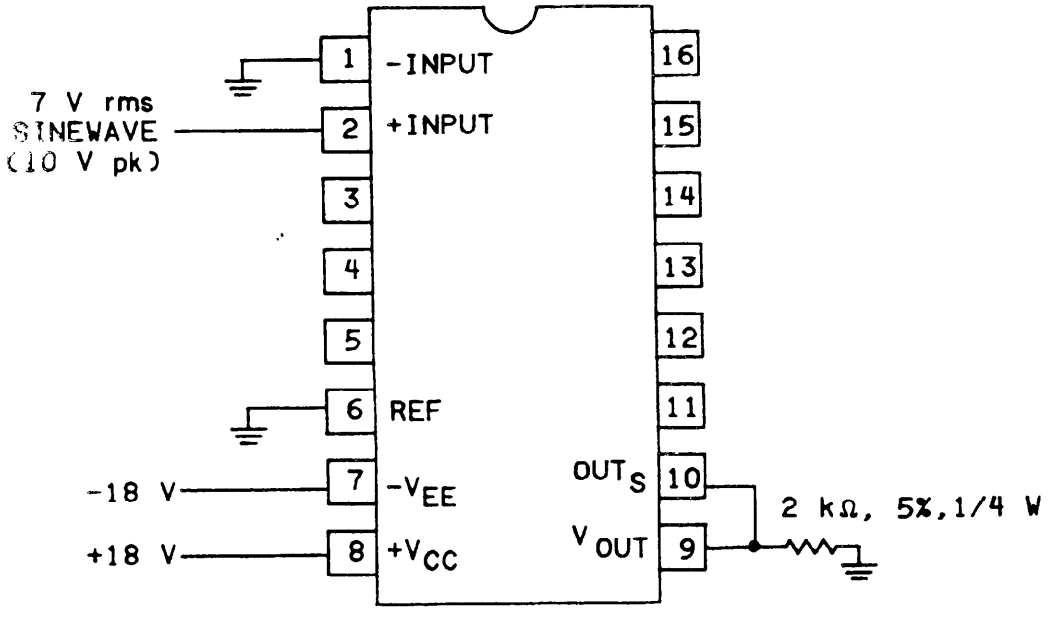
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Parameter symbol	Gain	Input pulse signal at $t_r \leq 50$ ns	Output pulse signal	Equation
SR (+)	1	+10 V to -10 V step	Waveform 1	$SR(+) = \Delta V_O(+)/\Delta t(+)$
SR (-)	1	-10 V to +10 V step	Waveform 2	$SR(-) = \Delta V_O(-)/\Delta t(-)$
SR (+)	1000	+10 mV to -10 mV step	Waveform 1	$SR(+) = \Delta V_O(+)/\Delta t(+)$
SR (-)	1000	-10 mV to +10 mV step	Waveform 2	$SR(-) = \Delta V_O(-)/\Delta t(-)$

FIGURE 9. Waveforms for slew rate.

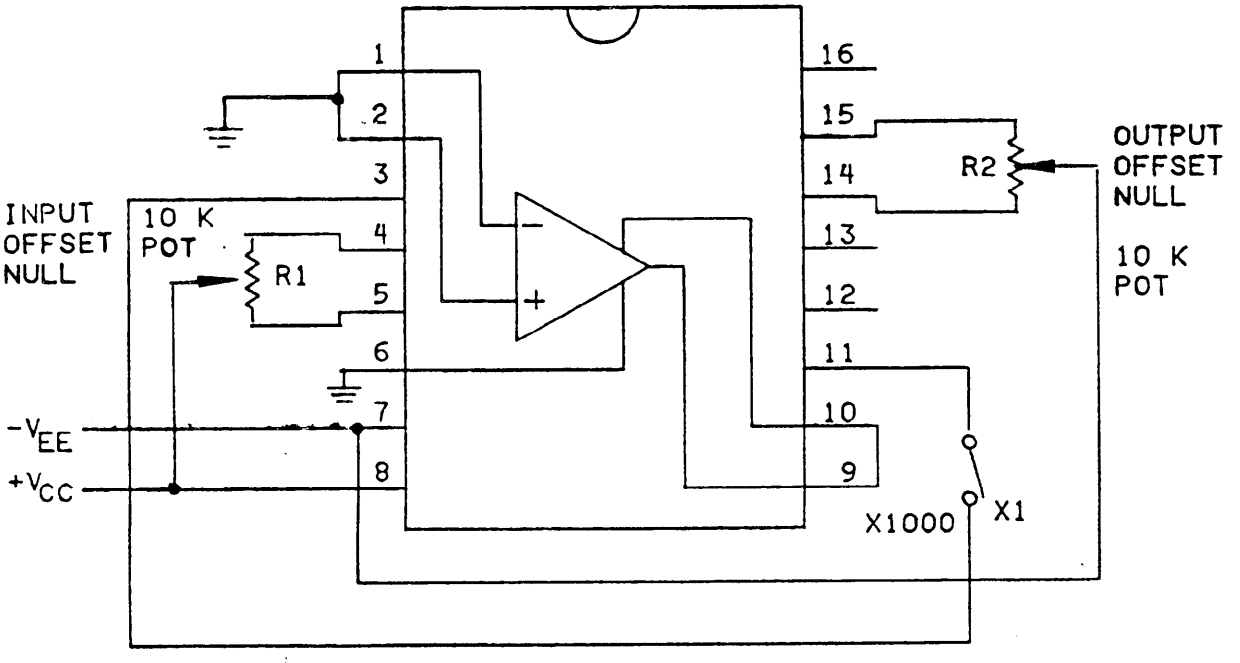
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NOTE: Burn-in boards shall be constructed with adequate capacitive decoupling of the ±18 V busses to eliminate device oscillations.

FIGURE 10. Test circuit for burn-in and steady-state life test.

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Input and output offset adjustment procedure:

1. All adjustments are made at +25°C with both inverting and noninverting inputs grounded. Care must be taken to prevent oscillations at high gains.
2. To null the input offset, select the gain A = 1000.
3. Trim R1 to null the output.
4. To null the output offset, select the gain A = 1.
5. Trim R2 to null the output.
6. Repeat steps 2-5 iteratively until no change is observed at the output.

FIGURE 11. Offset null circuit.

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TABLE III. Group A inspection.

Subgroup	Symbol	Test no.	MIL-STD-883 method	Adapter setup			Energized relays	Measured pin			DUT Gain (A) Equations	Notes	Limits		Unit
				$+V_{CC}$	V_{IN}	Gain	(Figure 3)	Number	Value	Unit			Min	Max	
1 T_A = +25°C	I_{CC}	1	4005	±15	0	1	K1, K5, K12	8	I1	mA	A = 1 I_{CC} = I1		0	5	mA
	I_{EE}	2	4005	±15	0	1	K1, K5, K12	7	I2	mA	A = 1 I_{EE} = I2		-5	0	mA
	$I_{OS}(+)$	3	3011	±15	0	1	K1, K5, K12	2	I3	mA	A = 1 $I_{OS}(+)$ = I3	1/	10	85	mA
	$I_{OS}(-)$	4	3011	±15	0	1	K1, K5, K12	2	I4	mA	A = 1 $I_{OS}(-)$ = I4	1/	-85	-10	mA
	V_{IOS}	5	4001	±15	0	100	K5, K12, K16	1	E1	V	A = 1	2/ 3/			
							K1, K5, K12, K16	1	E2	V	A = 1				
						10	K4, K5, K12, K15	1	E3	V	A = 1000	2/			
							K1, K4, K5, K12, K15	1	E4	V	A = 1000 $V_{IOS} = \frac{(E4-F3)-(E2-E1)}{.00999 \quad .0999}$		-100	100	µV
	V_{OOS}	6	4001								$V_{OOS} = \frac{(E2-E1)-(E4-E3)}{.0999 \quad 9.99}$		-3	3	mV
	$V_{IOSAdj}(+)$	7	4001	±15	0	1	K4, K5, K12, K17, K24	1	E5	mV	A = 1000 $V_{IOSAdj}(+) = (E5-(V_{IOS}*V_{OOS}))$		103		µV
	$V_{IOSAdj}(-)$	8	4001	±15	0	1	K4, K5, K12, K18, K20, K24	1	E6	mV	A = 1000 $V_{IOSAdj}(-) = (E6-(V_{IOS}*V_{OOS}))$			-103	µV
	$V_{OOSAdj}(+)$	9	4001	±15	0	1	K5, K12, K18, K24	1	E7	mV	A = 1 $V_{OOSAdj}(+) = (E7-(V_{OOS}*(V_{IOS}/1000))$		3.1		mV
	$V_{OOSAdj}(-)$	10	4001	±15	0	1	K5, K12, K18, K19, K24	1	E8	mV	A = 1 $V_{OOSAdj}(-) = (E8-(V_{OOS}*(V_{IOS}/1000))$			-3.1	mV
	$I_{IB}(+)$	11	4001	±15	0	100	K1, K5, K12, K16	1	E9	V	A = 1				
							K1, K7, K12, K16	1	E10	V	A = 1 $I_{IB}(+)$ = 100(E10-E9)		-50	50	nA
	$I_{IB}(-)$	12	4001	±15	0	100	K1, K5, K16	1	E11	V	A = 1 $I_{IB}(-)$ = 100(E11-E9)		-50	50	nA
	I_{IO}	13	4001								$I_{IO} = I_{IB}(+)-I_{IB}(-)$		-35	35	nA
	P_{SRRA}	14	4003	±16.5	0	100	K1, K5, K12, K16	1	E12	V	A = 1	3/			
				±10.8	0	100	K1, K5, K12, K16	1	E13	V	A = 1 $P_{SRRA} = 20LOG\frac{(1140)}{(E12-E13)}$		75		dB
	P_{SRRB}	15	4003	±16.5	0	100	K1, K2, K5, K12, K16	1	E14	V	A = 10				
				±10.8	0	100	K1, K2, K5, K12, K16	1	E15	V	A = 10 $P_{SRRB} = 20LOG\frac{(11400)}{(E14-F15)}$		95		dB

See footnotes at end of table.

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TABLE III. Group A inspection - Continued.

Subgroup	Symbol	Test no.	MIL-STD-883 method	Adapter setup			Energized relays	Measured pin			DUT Gain (A) Equations	Notes	Limits		Unit
				$\pm V_{CC}$	V_{IN}	Gain	(Figure 3)	Number	Value	Unit			Min	Max	
1 T_A = +25°C	P_{SRRC}	16	4003	±16.5	0	100	K1, K3, K5, K12, K16	1	E16	V	A = 100				
				±10.8	0	100	K1, K3, K5, K12, K16	1	E17	V	A = 100 $P_{SRRC} = 20LOG\frac{(114000)}{(E16-E17)}$		105		dB
	P_{SRRD}	17	4003	±16.5	0	10	K1, K4, K5, K12, K16	1	E18	V	A = 1000				
				±10.8	0	10	K1, K4, K5, K12, K16	1	E19	V	A = 1000 $P_{SRRD} = 20LOG\frac{(1140000)}{(E18-E19)}$		110		dB
	C_{MRRA} (+)	18	4003	±15	+10	100	K1, K13, K14, K16	1	E20	V	A = 1	3/			
					0	100	K1, K13, K14, K16	1	E21	V	A = 1 $C_{MRRA}(+) = 20LOG\frac{(1000)}{(E20-E21)}$	4/	70		dB
	C_{MRRA} (-)	19	4003	±15	-10	100	K1, K13, K14, K16	1	E22	V	A = 1 $C_{MRRA}(-) = 20LOG\frac{(1000)}{(E21-E22)}$	4/	70		dB
	C_{MRRB} (+)	20	4003	±15	+10	100	K1, K2, K13, K14, K16	1	E23	V	A = 10				
					0	100	K1, K2, K13, K14, K16	1	E24	V	A = 10 $C_{MRRB}(+) = 20LOG\frac{(10000)}{(E23-E24)}$	4/	90		dB
	C_{MRRB} (-)	21	4003	±15	-10	100	K1, K2, K13, K14, K16	1	E25	V	A = 10 $C_{MRRB}(-) = 20LOG\frac{(10000)}{(E24-E25)}$	4/	90		dB
				Adapter supplies											
	C_{MRRC} (+)	22	4003	±15	+10	100	K1, K3, K13, K14, K16	1	E26	V	A = 100				
					0	100	K1, K3, K13, K14, K16	1	E27	V	A = 100 $C_{MRRC}(+) = 20LOG\frac{(100000)}{(E26-E27)}$	4/	100		dB
	C_{MRRC} (-)	23	4003	±15	-10	100	K1, K3, K13, K14, K16	1	E28	V	A = 100 $C_{MRRC}(-) = 20LOG\frac{(100000)}{(E27-E28)}$	4/	100		dB
	C_{MRRD} (+)	24	4003	±15	+10	10	K1, K4, K13, K14, K15	1	E29	V	A = 1000	2/ 3/			
					0	10	K1, K4, K13, K14, K15	1	E30	V	A = 1000 $C_{MRRD}(+) = 20LOG\frac{(1000000)}{(E29-E30)}$	4/	110		dB
	C_{MRRD} (-)	25	4003	±15	-10	10	K1, K4, K13, K14, K15	1	E31	V	A = 1000 $C_{MRRD}(-) = 20LOG\frac{(1000000)}{(E30-E31)}$	4/	110		dB

See footnotes at end of table.

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TABLE III. Group A inspection - Continued

Subgroup	Symbol	Test no.	MIL-STD-883 method	Adapter setup			Energized relays	Measured pin			DUT Gain (A) equations	Notes	Limits		Unit
				+V_{CC}	V_{IN}	Gain	(Figure 3)	Number	Value	Unit			Min	Max	
2 T_A = +125°C	I_{CC}	26	4005	±15	0	1	K1, K5, K12	8	I5	mA	A = 1 I_{CC} = I5		0	5	mA
	I_{EE}	27	4005	±15	0	1	K1, K5, K12	7	I6	mA	A = 1 I_{EE} = I6		-5	0	mA
	$I_{OS(+)}$	28	3011	±15	0	1	K1, K5, K12	2	I7	mA	A = 1 $I_{OS(+)}$ = I7	1/	10	85	mA
	$I_{OS(-)}$	29	3011	±15	0	1	K1, K5, K12	2	I8	mA	A = 1 $I_{OS(-)}$ = I8	1/	-85	-10	mA
	V_{IOS}	30	4001	±15	0	100	K5, K12, K16	1	E32	V	A = 1	2/ 3/			
							K1, K5, K12, K16	1	E33	V	A = 1				
						10	K4, K5, K12, K15	1	E34	V	A = 1000	2/			
							K1, K4, K5, K12, K15	1	E35	V	A = 1000 $V_{IOS} = \frac{(E35-E34)-(E33-E32)}{.00999 \quad .0999}$		-300	300	µV
	dV_{IOS}/dT	31	4001	±15	0	0					dV_{IOS}/dT = (V_{IOS} at +125°C-V_{IOS} at +25°C/100)	3/ 5/	-2	2	µV/°C
	V_{OOS}	32	4001								$V_{OOS} = \frac{(E33-E32)-(E35-E34)}{.0999 \quad 9.99}$		-8	8	mV
	dV_{OOS}/dT	33	4001								dV_{OOS}/dT = 10(V_{OOS} at +125°C-V_{OOS} at +25°C)		-50	50	µV/°C
	$I_{IB(+)}$	34	4001	±15	0	100	K1, K5, K12, K16	1	E36	V	A = 1				
							K1, K7, K12, K16	1	E37	V	A = 1 $I_{IB(+)}$ = 100(E37-E36)		-70	70	nA
	$I_{IB(-)}$	35	4001	±15	0	100	K1, K5, K16	1	E38	V	A = 1 $I_{IB(-)}$ = 100(E38-E36)		-70	70	nA
	I_{IO}	36	4001								I_{IO} = $I_{IB(+)}$-$I_{IB(-)}$		-50	50	nA
	P_{SRRA}	37	4003	±16.5	0	1	K1, K5, K12, K16	100	E39	V	A = 1	3/			
				±10.8	0	1	K1, K5, K12, K16	100	E40	V	A = 1 $P_{SRRA} = 20LOG\frac{(1140)}{(E39-E40)}$		75		dB
	P_{SRRB}	38	4003	±16.5	0	100	K1, K2, K5, K12, K16	1	E41	V	A = 10				
				±10.8	0	100	K1, K2, K5, K12, K16	1	E42	V	A = 10 $P_{SRRB} = 20LOG\frac{(11400)}{(E42-E41)}$		95		dB
	P_{SRRC}	39	4003	±16.5	0	100	K1, K3, K5, K12, K16	1	E43	V	A = 100				
				±10.8	0	100	K1, K3, K5, K12, K16	1	E44	V	A = 100 $P_{SRRC} = 20LOG\frac{(114000)}{(E44-E43)}$		105		dB

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TABLE III. Group A inspection - Continued.

Subgroup	Symbol	Test no.	MIL-STD-883 method	Adapter setup			Energized relays	Measured pin			DUT Gain (A) Equations	Notes	Limits		Unit
				$+V_{CC}$	V_{IN}	Gain	(Figure 3)	Number	Value	Unit			Min	Max	
2 T_A = +125°C	P_{SRRD}	40	4003	+16.5	0	10	K1, K4, K5, K12, K15	1	E45	V	A = 1000				
				+10.8	0	10	K1, K4, K5, K12, K15	1	E46	V	A = 1000 $P_{SRRD} = 20LOG\frac{(1140000)}{(E46-E45)}$		110		dB
	$C_{MRRA(+)}$	41	4003	+15	+10	100	K1, K13, K14, K16	1	E47	V	A = 1	3/			
					0	100	K1, K13, K14, K16	1	E48	V	A = 1 $C_{MRRA(+)} = 20LOG\frac{(1000)}{(E47-E48)}$	4/	70		dB
	$C_{MRRA(-)}$	42	4003	+15	-10	100	K1, K13, K14, K16	1	E49	V	A = 1 $C_{MRRA(-)} = 20LOG\frac{(1000)}{(E49-E50)}$	4/	70		dB
	$C_{MRRB(+)}$	43	4003	+15	+10	100	K1, K2, K13, K14, K16	1	E50	V	A = 10				
					0	100	K1, K2, K13, K14, K16	1	E51	V	A = 10 $C_{MRRB(+)} = 20LOG\frac{(10000)}{(E51-E52)}$	4/	90		dB
	$C_{MRRB(-)}$	44	4003	+15	-10	100	K1, K2, K13, K14, K16	1	E52	V	A = 10 $C_{MRRB(-)} = 20LOG\frac{(10000)}{(E53-E54)}$	4/	90		dB
				Adapter supplies											
	$C_{MRRC(+)}$	45	4003	+15	+10	100	K1, K3, K13, K14, K16	1	E53	V	A = 100				
					0	100	K1, K3, K13, K14, K16	1	E54	V	A = 100 $C_{MRRC(+)} = 20LOG\frac{(100000)}{(E53-E54)}$	4/	100		dB
	$C_{MRRC(-)}$	46	4003	+15	-10	100	K1, K3, K13, K14, K16	1	E55	V	A = 100 $C_{MRRC(-)} = 20LOG\frac{(100000)}{(E54-E55)}$	4/	100		dB
	$C_{MRRD(+)}$	47	4003	+15	+10	10	K1, K4, K13, K14, K15	1	E56	V	A = 1000	2/ 3/			
					0	10	K1, K4, K13, K14, K15	1	E57	V	A = 1000 $C_{MRRD(+)} = 20LOG\frac{(1000000)}{(E56-E57)}$	4/	110		dB
	$C_{MRRD(-)}$	48	4003	+15	-10	10	K1, K4, K13, K14, K15	1	E58	V	A = 1000 $C_{MRRD(-)} = 20LOG\frac{(1000000)}{(E57-E58)}$	4/	110		dB

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Subgroup	Symbol	Test no.	MIL-STD-883 method	Adapter setup			Energized relays	Measured pin			DUT Gain (A) Equations	Notes	Limits		Unit
				$+V_{CC}$	V_{IN}	Gain	(Figure 3)	Number	Value	Unit			Min	Max	
3 T_A = -55°C	I_{CC}	49	4005	±15	0	1	K1, K5, K12	8	19	mA	A = 1 I_{CC} = 19		0	5	mA
	I_{EE}	50	4005	±15	0	1	K1, K5, K12	7	110	mA	A = 1 I_{EE} = 110		-5	0	mA
	$I_{OS(+)}$	51	3011	±15	0	1	K1, K5, K12	2	111	mA	A = 1 $I_{OS(+)}$ = 111	1/	10	85	nA
	$I_{OS(-)}$	52	3011	±15	0	1	K1, K5, K12	2	112	mA	A = 1 $I_{OS(-)}$ = 112	1/	-85	-10	nA
	V_{IOS}	53	4001	±15	0	100	K5, K12, K16	1	E59	V	A = 1	2/ 3/			
							K1, K5, K12, K16	1	E60	V	A = 1				
						10	K4, K5, K12, K15	1	E61	V	A = 1000	2/			
							K1, K4, K5, K12, K15	1	E62	V	A = 1000 $V_{IOS} = \frac{(E62-E61)-(E60-E59)}{.00999 \quad .0999}$		-260	260	µV
	dV_{IOS}/dT	54	4001	±15	0	0					$dV_{IOS}/dT = (V_{IOS}$ at +25°C-V_{IOS} at -55°C)/80	3/ 5/	-2	2	µV/°C
	V_{OOS}	55	4001								$V_{OOS} = \frac{(E60-E59)-(E62-E61)}{.0999 \quad 9.99}$		-7	7	mV
	dV_{OOS}/dT	56	4001								dV_{OOS}/dT = 12.5(V_{OOS} at +25°C-V_{OOS} at -55°C)		-50	50	µV/°C
	$I_{IB(+)}$	57	4001	±15	0	100	K1, K5, K12, K16	1	E63	V	A = 1				
							K1, K7, K12, K16	1	E64	V	A = 1 $I_{IB(+)}$ = 100(E64-E63)		-70	70	nA
	$I_{IB(-)}$	58	4001	±15	0	100	K1, K5, K16	1	E65	V	A = 1 $I_{IB(-)}$ = 100(E65-E63)		-70	70	nA
	I_{IO}	59	4001								$I_{IO} = I_{IB(+)}-I_{IB(-)}$		-50	50	nA
	P_{SRRA}	60	4003	±16.5	0	100	K1, K5, K12, K16	1	E66	V	A = 1	3/			
				±10.8	0	100	K1, K5, K12, K16	1	E67	V	A = 1 $P_{SRRA} = 20LOG\frac{(1140)}{(E66-E67)}$		75		dB
	P_{SRRB}	61	4003	±16.5	0	100	K1, K2, K5, K12, K16	1	E68	V	A = 10				
				±10.8	0	100	K1, K2, K5, K12, K16	1	E69	V	A = 10 $P_{SRRB} = 20LOG\frac{(11400)}{(E68-E69)}$		95		dB

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TABLE III. Group A inspection - Continued.

Subgroup	Symbol	Test no.	MIL-STD-883 method	Adapter setup			Energized relays	Measured pin			DUT Gain (A) Equations	Notes	Limits		Unit
				$\pm V_{CC}$	V_{IN}	Gain	(Figure 3)	Number	Value	Unit			Min	Max	
3 T_A = -55°C	P_{SRRC}	62	4003	±16.5	0	100	K1, K3, K5, K12, K16	1	E70	V	A = 100				
				±10.8	0	100	K1, K3, K5, K12, K16	1	E71	V	A = 100 $P_{SRRC} = 20LOG\frac{(114000)}{(E70-E71)}$		105		dB
	P_{SRRD}	63	4003	±16.5	0	10	K1, K4, K5, K12, K15	1	E72	V	A = 1000				
				±10.8	0	10	K1, K4, K5, K12, K15	1	E73	V	A = 1000 $P_{SRRD} = 20LOG\frac{(1140000)}{(E72-E73)}$		110		dB
	$C_{MRRA(+)}$	64	4003	±15	+10	100	K1, K13, K14, K16	1	E74	V	A = 1	3/			
					0	100	K1, K13, K14, K16	1	E75	V	A = 1 $C_{MRRA(+)} = 20LOG\frac{(1000)}{(E74-E75)}$	4/	70		dB
	$C_{MRRA(-)}$	65	4003	±15	-10	100	K1, K13, K14, K16	1	E76	V	A = 1 $C_{MRRA(-)} = 20LOG\frac{(1000)}{(E75-E76)}$	4/	70		dB
	$C_{MRRB(+)}$	66	4003	±15	+10	100	K1, K2, K13, K14, K16	1	E77	V	A = 10				
					0	100	K1, K2, K13, K14, K16	1	E78	V	A = 10 $C_{MRRB(+)} = 20LOG\frac{(10000)}{(E77-E78)}$	4/	90		dB
	$C_{MRRB(-)}$	67	4003	±15	-10	100	K1, K2, K13, K14, K16	1	E79	V	A = 10 $C_{MRRB(-)} = 20LOG\frac{(10000)}{(E78-E79)}$	4/	90		dB
				Adapter supplies											
	$C_{MRRC(+)}$	68	4003	±15	+10	100	K1, K3, K13, K14, K16	1	E80	V	A = 100				
					0	100	K1, K3, K13, K14, K16	1	E81	V	A = 100 $C_{MRRC(+)} = 20LOG\frac{(100000)}{(E80-E81)}$	4/	100		dB
	$C_{MRRC(-)}$	69	4003	±15	-10	100	K1, K3, K13, K14, K16	1	E82	V	A = 100 $C_{MRRC(-)} = 20LOG\frac{(100000)}{(E81-E82)}$	4/	100		dB
	$C_{MRRD(+)}$	70	4003	±15	+10	10	K1, K4, K13, K14, K15	1	E83	V	A = 1000	2/ 3/			
					0	10	K1, K4, K13, K14, K15	1	E84	V	A = 1000 $C_{MRRD(+)} = 20LOG\frac{(1000000)}{(E83-E84)}$	4/	110		dB
	$C_{MRRD(-)}$	71	4003	±15	-10	10	K1, K4, K13, K14, K15	1	E85	V	A = 1000 $C_{MRRD(-)} = 20LOG\frac{(1000000)}{(E84-E85)}$	4/	110		dB

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TABLE III. Group A inspection - Continued.

Subgroup	Symbol	Test no.	MIL-STD-883 method	Adapter			Energized relays	Measured pin			DUT Gain (A) equations	Notes	Limits		Unit
				±V_{CC}	V_{IN}	Gain	(Figure 3)	Number	Value	Unit			Min	Max	
4 T_A = +25°C	A_{EA}	72	4004	±15	+10	100	K1, K8, K12, K16, K23	1	E86	V	A = 1				
					-10	100	K1, K8, K12, K16, K23	1	E87	V	A = 1 $A_{EA} = \frac{(E86-E87)}{20}$		-0.05	0.05	%
	A_{EB}	73	4004	±15	+10	100	K1, K2, K6, K9, K12, K16, K23	1	E88	V	A = 10				
					+10	100	K1, K2, K6, K9, K12, K16, K23	1	E89	V	A = 10 $A_{EB} = \frac{(E88-E89)}{20}$		-0.25	0.25	%
	A_{EC}	74	4004	±15	+10	100	K1, K3, K6, K10, K12, K16, K23	1	E90	V	A = 100				
					-10	100	K1, K3, K6, K10, K12, K16, K23	1	E91	V	A = 100 $A_{EC} = \frac{(E90-E91)}{20}$		-0.50	0.50	%
	A_{ED}	75	4004	±15	+10	10	K1, K4, K6, K11, K12, K15, K23	1	E92	V	A = 1000				
					-10	10	K1, K4, K6, K11, K12, K15, K23	1	E93	V	A = 1000 $A_{ED} = \frac{(E92-E93)}{2}$		-2.0	2.0	%
	$V_{OP}(+)$	76	4004	±15	+15	1	K12, K14, K22, K24	1	E94	V	A = 1 $V_{OP}(+) = E94$	3/ 5/	10	15	V
	$V_{OP}(-)$	77	4004	±15	-15	1	K5, K13, K22, K24	1	E95	V	A = 1 $V_{OP}(-) = E95$		-15	-10	V
5 T_A = +125°C	dA_{EA}/dT	78	4004	±15	+10	100	K1, K8, K12, K16, K23	1	E96	V	A = 1				
					-10	100	K1, K8, K12, K16, K23	1	E97	V	A = 1 $dA_{EA}/dT = \frac{((E96-E97)-(E86-E87))10}{2}$		-5	5	ppm/°C
	dA_{EB}/dT	79	4004	±15	+10	100	K1, K2, K6, K9, K12, K16, K23	1	E98	V	A = 10				
					-10	100	K1, K2, K6, K9, K12, K16, K23	1	E99	V	A = 10 $dA_{EB}/dT = \frac{((E98-E99)-(E88-E89))10}{2}$		-10	10	ppm/°C
	dA_{EC}/dT	80	4004	±15	+10	100	K1, K3, K6, K10, K12, K16, K23	1	E100	V	A = 100				
					-10	100	K1, K3, K6, K10, K12, K16, K23	1	E101	V	A = 100 $dA_{EC}/dT = \frac{((E100-E101)-(E90-E91))10}{2}$		-25	25	ppm/°C

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TABLE III. Group A inspection - Continued.

Subgroup	Symbol	Test no.	MIL-STD-883 method	Adapter			Energized relays	Measured pin			DUT Gain (A) Equations	Notes	Limits		Unit
				$+V_{CC}$	V_{IN}	Gain	(Figure 3)	Number	Value	Unit			Min	Max	
5 T_A = +125°C	dA_{ED}/dT	81	4004	±15	+10	10	K1, K4, K6, K11, K12, K15, K23	1	E102	V	A = 1000				
					-10	10	K1, K4, K6, K11, K12, K15, K23	1	E103	V	A = 1000 $dA_{ED}/dT = \frac{((E102-E103)-(E92-E93))10^2}{2}$		-50	50	ppm/°C
	$V_{OP}(+)$	82	4004	±15	+15	1	K12, K14, K22, K24	1	E104	V	A = 1 $V_{OP}(+)$ = E104	3/ 5/	10	15	V
	$V_{OP}(-)$	83	4004	±15	-15	1	K5, K13, K22, K24	1	E105	V	A = 1 $V_{OP}(-)$ = E105		-15	-10	V
6 T_A = -55°C	dA_{EA}/dT	84	4004	±15	+10	100	K1, K8, K12, K16, K23	1	E106	V	A = 1				
					-10	100	K1, K8, K12, K16, K23	1	E107	V	A = 1 $dA_{EA}/dT = \frac{((E86-E87)-(E106-E107))10^2}{16}$		-5	5	ppm/°C
	dA_{EB}/dT	85	4004	±15	+10	100	K1, K2, K6, K9, K12, K16, K23	1	E108	V	A = 10				
					-10	100	K1, K2, K6, K9, K12, K16, K23	1	E109	V	A = 10 $dA_{EB}/dT = \frac{((E88-E89)-(E108-E109))10^2}{16}$		-10	10	ppm/°C
	dA_{EC}/dT	86	4004	±15	+10	100	K1, K3, K6, K10, K12, K16, K23	1	E110	V	A = 100				
					-10	100	K1, K3, K6, K10, K12, K16, K23	1	E111	V	A = 100 $dA_{EC}/dT = \frac{((E90-E91)-(E110-E111))10^2}{16}$		-25	25	ppm/°C
	dA_{ED}/dT	87	4004	±15	+10	10	K1, K4, K6, K11, K12, K15, K23	1	E112	V	A = 1000				
					-10	10	K1, K4, K6, K11, K12, K15, K23	1	E113	V	A = 1000 $dA_{ED}/dT = \frac{((E92-E93)-(E112-E113))10^3}{16}$		-50	50	ppm/°C
	$V_{OP}(+)$	88	4004	±15	+15	1	K12, K14, K22, K24	1	E114	V	A = 1 $V_{OP}(+)$ = E114	3/ 5/	10	15	V
	$V_{OP}(-)$	89	4004	±15	-15	1	K5, K13, K22, K24	1	E115	V	A = 1 $V_{OP}(-)$ = E115		-15	-10	V

See footnotes at end of table.

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TABLE III. Group A inspection - Continued.

Subgroup	Symbol	Test no.	MIL-STD-883 method	Adapter +V_{CC}	Adapter V_{IH}	Adapter Gain	Switch positions	Measured pin Number	Measured pin Value	Measured pin Unit	OUT Gain (A) Equations and figures	Notes	Limits Min	Limits Max	Unit
7 T_A = +25°C	ts1			±15			S1 = 1X S2 = 1X	1	t1	V	A = 1 ts1 = t1 Figures 4, 7			15	µs
	ts2			±15			S1 = 10X S2 = 10X	1	t2	V	A = 10 ts2 = t2 Figures 4, 7			15	µs
	ts3			±15			S1 = 100X S2 = 100X	1	t3	V	A = 100 ts3 = t3 Figures 4, 7			15	µs
	ts4			±15			S1 = 1000X S2 = 1000X	1	t4	V	A = 1000 ts4 = t4 Figures 4, 7			75	µs
	SR1(+)			±15			S1 OPEN	1	t5	V	A = 1 SR1(+) = t5 Figures 8, 9		3.5		V/µs
	SR1(-)			±15			S1 OPEN	1	t6	V	A = 1 SR1(-) = t6 Figures 8, 9			-3.5	V/µs
	SR2(+)			±15			S1 CLOSED	1	t7	V	A = 1000 SR2(+) = t7 Figures 8, 9		0.5		V/µs
	SR2(-)			±15			S1 CLOSED	1	t8	V	A = 1000 SR2(-) = t8 Figures 8, 9			-0.5	V/µs
	En			±15				1	E116	V rms	A = 1 En = E116 Figure 5		0	350	$nV/\sqrt{Hz}$
	In			±15				1	I13	µA rms	A = 1 In = I13 Figure 5		0	10	$pA/\sqrt{Hz}$
	Enpp			±15				1	E117	V_{p-p}	A = 1 Enpp = E117/20000 Figure 6		0	15	μV_{p-p}

1/ $I_{OS}(+)$ and $I_{OS}(-)$ are measured with the output shorted. The duration of the output short-circuit current shall not cause the junction temperature to exceed its maximum rating.
2/ This is a gain amplifier offset measurement at the specified gain.
3/ Unless otherwise specified, all tests apply to figure 3. The gain amplifier shall have a separate power supply with respect to the DUT power supply.
4/ C_{MRR} is referenced to the output (RTO).
5/ XXX/dT is calculated using data from previous tests.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Burn-in tests (method 1015 of MIL-STD-883).

(1) For class S devices, test condition D, using the circuit shown on figure 10.

(2) For class B devices, test condition D, using the circuit shown on figure 10.

b. Interim and final electrical test parameters shall be as specified in table II, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

c. The percent defective allowable (PDA) for class S and class B devices shall be as specified in MIL-M-38510, based on failures from group A, subgroup 1 test after cooldown as final electrical test in accordance with method 5004 of MIL-STD-883 and with no intervening electrical measurements. If interim electrical parameter tests are performed prior to burn-in, failures resulting from pre burn-in screening may be excluded from the PDA. If interim electrical parameter tests prior to burn-in are omitted, then all screening failures shall be included in the PDA. The verified failures of group A, subgroup 1 after burn-in divided by the total number of devices submitted for burn-in in that lot shall be used to determine the percent defective for that lot, and the lot shall be accepted or rejected based on the PDA for the applicable device class.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Tests shall be as specified in table II herein.

b. Subgroups 8, 9, 10, and 11 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein. For class S devices, delta limits shall apply only to subgroup 5 of group B inspection.

b. Steady-state life test for class S devices shall be in accordance with table IIa of method 5005 of MIL-STD-883, using a circuit shown on figure 10. If the alternate burn-in conditions are used, the circuit shown on figure 10 shall be used.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein. Delta limits shall apply only to subgroup 1 of group C inspection for class B devices.

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b. Steady-state life test (method 1005 of MIL-STD-883) conditions:

(1) Test condition D using the circuit shown on figure 10.

(2) T_A = +125°C minimum.

(3) Test duration: 1,000 hours, except as permitted by appendix B of MIL-M-38510 and method 1005 of MIL-STD-883.

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical parameters shall be as specified in table II herein.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Voltage and current. All voltage values are referenced to the external zero reference level of the supply voltage. Currents given are conventional and positive when flowing into the referenced terminal.

4.5.2 Life test and burn-in cooldown procedure. When devices are measured at +25°C following application of the steady-state life or burn-in test condition, they shall be cooled within 10°C of their power stable condition at room temperature prior to removal of the bias.

4.6 Data reporting. When specified in the acquisition document, a copy of the following data, as applicable, shall be applied:

a. Attributes data for all screening tests (see 4.2) and variables data for all static burn-in, dynamic burn-in, and operating life tests.

b. A copy of each radiograph.

c. The quality conformance inspection data (see 4.4).

d. Parameter distribution data on parameters evaluated during burn-in (see 3.5).

e. Final electrical parameter data (see 4.2c).

TABLE IV. Group C end-point electrical parameters, T_A = +25°C.

Table III test number	Symbol	Delta		End-point limits		Unit
		Min	Max	Min	Max	
5	V_{IOS}	-75	75	-100	100	µV
6	V_{OOS}	-2	2	-3	3	mV
11	$I_{IB}(+)$	-25	25	-50	50	nA
12	$I_{IB}(-)$	-25	25	-50	50	nA
72	A_{EA}	-0.025	0.025	-0.05	0.05	%
73	A_{EB}	-0.125	0.125	-0.25	0.25	%
74	A_{EC}	-0.25	0.25	-0.5	0.5	%
75	A_{ED}	-1	1	-2	2	%

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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Intended use. Microcircuits conforming to this specification are intended for use for original equipment and logistic support of existing design application equipment.

6.2 Ordering data. Acquisition documents should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to the contracting activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action and reporting of results, if applicable.

f. Requirements for product assurance options.

g. Requirements for lead lengths or lead forming, if applicable, these requirements shall not affect the part number.

h. Requirements for "JAN" marking.

6.3 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510 and MIL-STD-1331.

6.4 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2), and lead material and finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.5 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-38510 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-38510 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	524S

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Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, MI
Navy - OS, SH
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Preparing activity:
Air Force - 17

Agent:
DLA - ES

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3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

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